

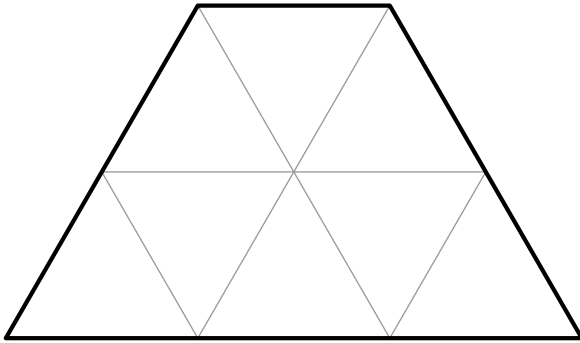
For every problem: each block must cover whole small triangles of the board, and blocks may not overlap or stick out past the edge of the board.

Problem 1: Cover each board with blue rhombuses, or decide that it cannot be done. When you cover a board, draw lines on it to show where your rhombuses went.



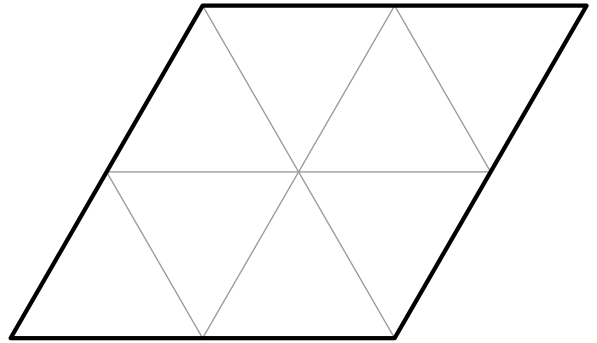
blue rhombuses

A



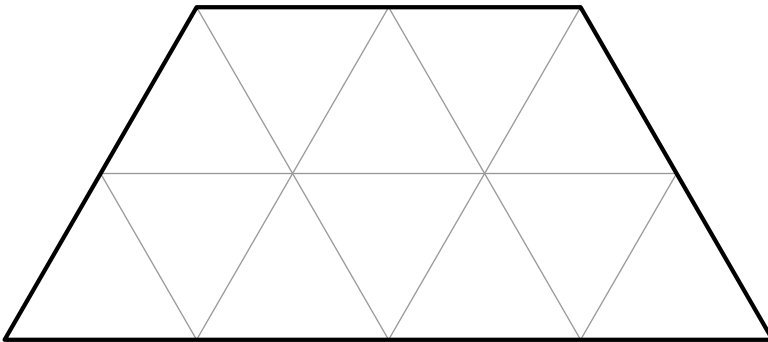
Can it be done? yes no

B



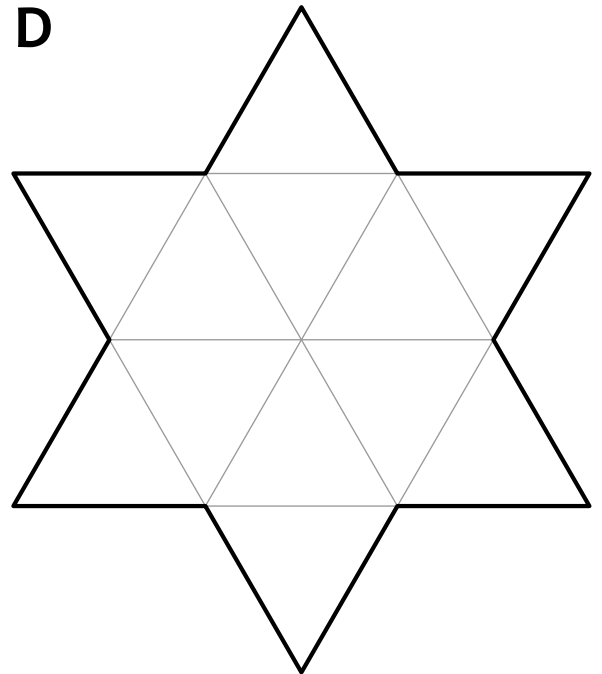
Can it be done? yes no

C



Can it be done? yes no

D



Can it be done? yes no

Problem 2: Cover each triangle board with blue rhombuses and green triangles, using as few green triangles as you can. Draw your best covering on the board, and write how many green triangles it used.

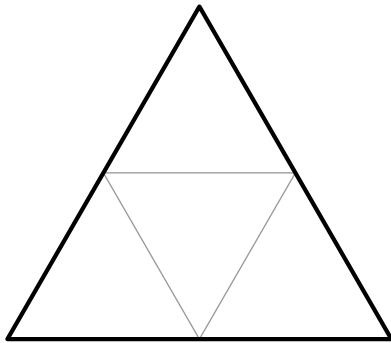


blue rhombuses



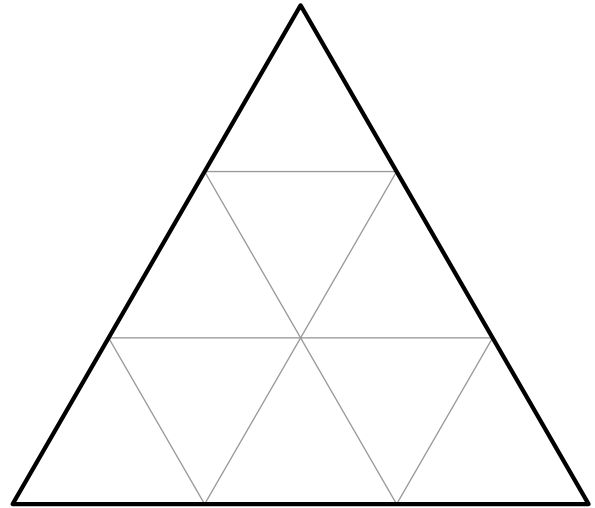
green triangles

E



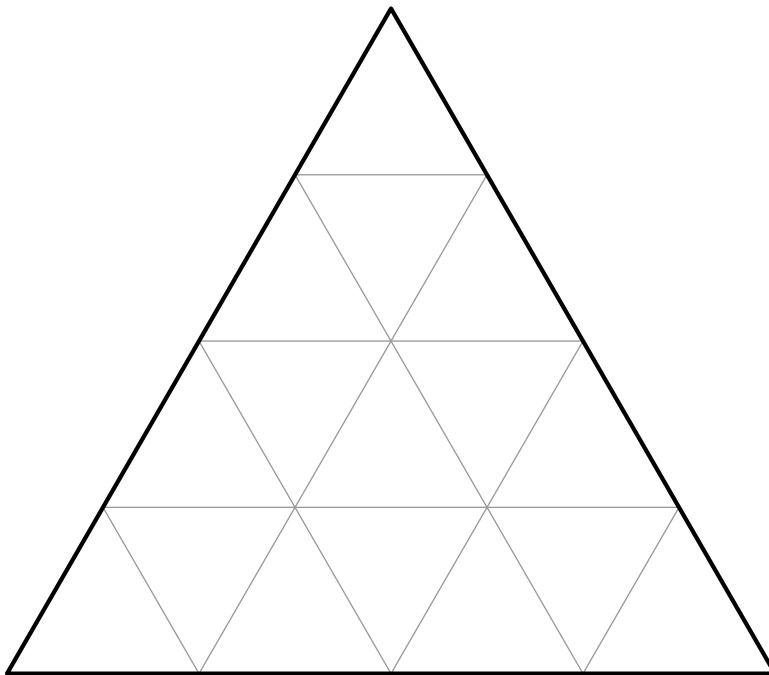
green triangles: _____

F



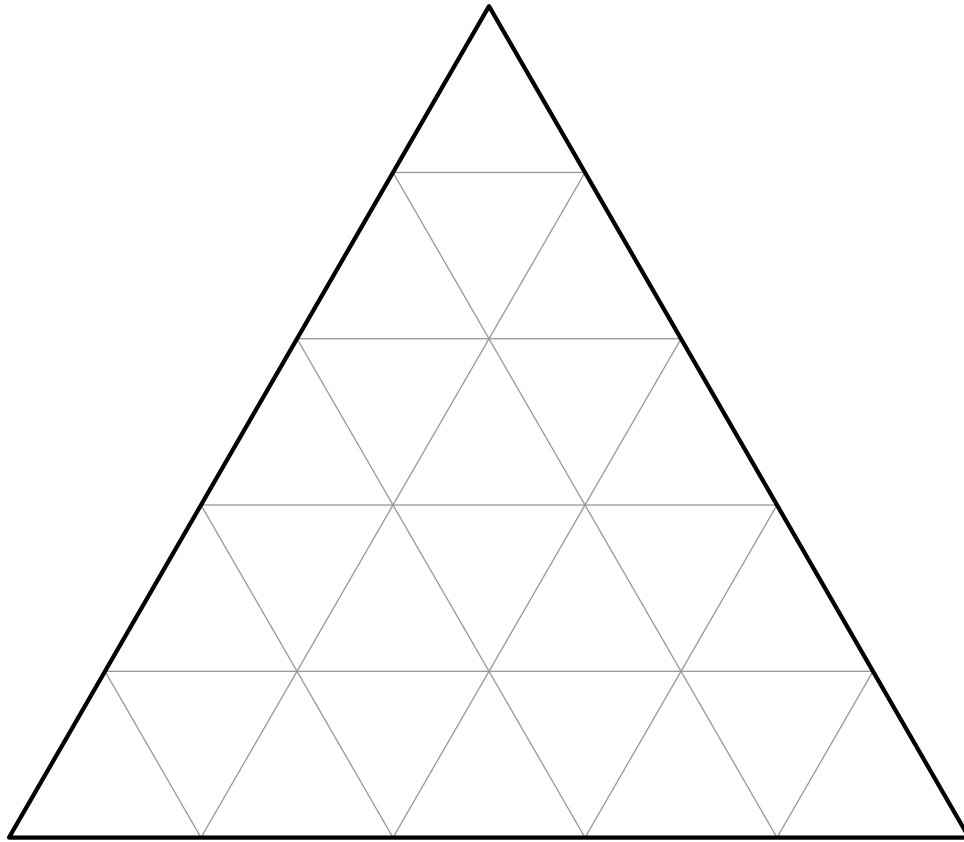
green triangles: _____

G



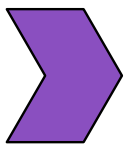
green triangles: _____

H



green triangles: _____

Problem 3: Now you may use purple chevrons as well as blue rhombuses and green triangles. Cover boards E, F, G, and H again. Can the chevrons help you use fewer green triangles than before? Explain.



purple chevrons



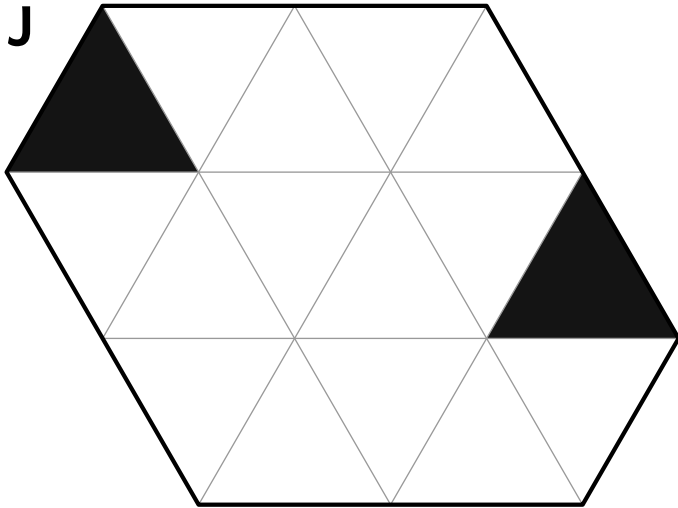
blue rhombuses



green triangles

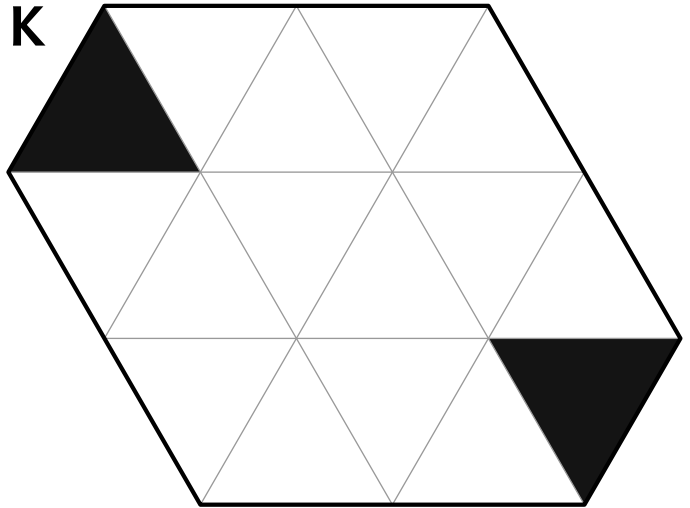
Problem 4: Each board below is the same hexagon with two small triangles colored black. Cover the white part with blue rhombuses, leaving the black triangles bare, or decide that it cannot be done. For each board that cannot be covered, explain why no one could ever cover it.

J



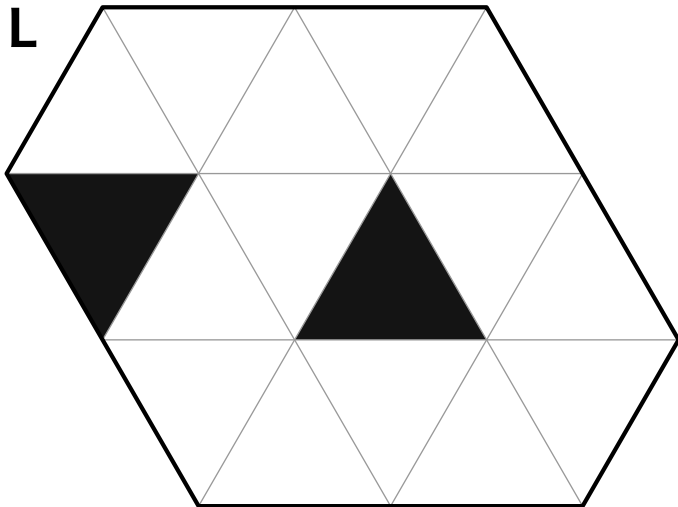
Can it be done? yes no

K



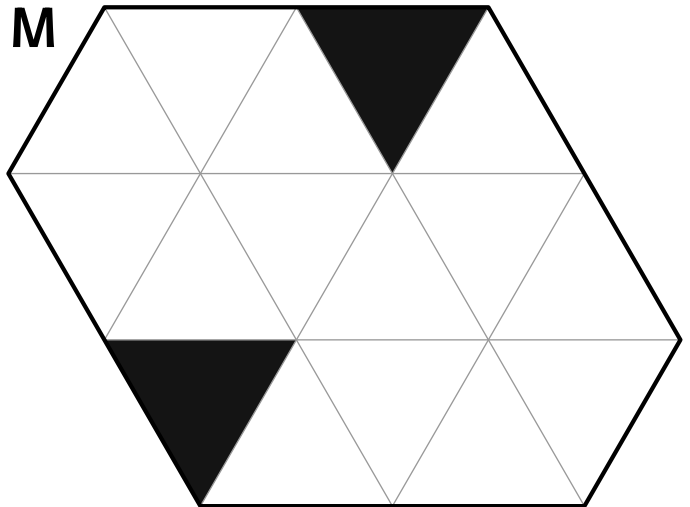
Can it be done? yes no

L



Can it be done? yes no

M



Can it be done? yes no

Problem 5: Explain why every covering of board G uses at least 4 green triangles, no matter where you put the blue rhombuses. Then imagine a triangle board with 10 small edges on each side. What is the fewest green triangles it needs? Explain.

Problem 6: On the grid below, draw a board made of exactly 16 small triangles, joined along their sides in one piece. Try to make a board that needs as many green triangles as possible when you cover it with blue rhombuses and green triangles. What is the most you can make it need?

